FW-eVTOL High-Fidelity 6DOF Flight Dynamics Simulation Framework

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Contents

1 List of Abbreviations

- u, v, w : Body-axis velocities (m/s)
- p, q, r : Body angular rates (rad/s)
- P_X, P_Y, P_Z : North-EAST-DOWN (NED) Position (m)
- q_0, q_1, q_2, q_3 : Quaternion components
- ϕ, θ, ψ : Euler angles (rad)
- D, L, Y: Drag, lift, side forces (N)
- L_a, M_a, N_a : Aerodynamic moments (N·m)
- $F_{X_p}, F_{Y_p}, F_{Z_p}$: Propulsion forces in body frame (N)
- L_p, M_p, N_p : Propulsion moments (N·m)
- ω_m : Motor mechanical angular velocity (rad/s)
- i_d, i_q : PMSM Stator currents (A)
- u_d, u_q : PMSM Stator voltages (V)
- $u_{turb}, v_{turb}, w_{turb}$: Dryden turbulence velocity components (m/s)
- $p_{turb}, q_{turb}, r_{turb}$: Angular turbulence velocity components (rad/s)
- $\sigma_u, \sigma_v, \sigma_w$: Turbulence intensity parameters (m/s)
- L_u, L_v, L_w : Turbulence length scales (m)
- m : Mass (kg)
- $I_{xx}, I_{yy}, I_{zz}, I_{xz}$: Inertia tensor elements (kg·m^2)
- Δ : $I_{xx}I_{zz} - I_{xz}^2$
- IMU: Inertial Measurement Unit
- $bias(t)$: bias process
- $noise(t)$: measurement noise
- $H(s)$: Sensor Transfer Function

2 Abstract

This project presents a comprehensive high-fidelity open-loop flight dynamics simulation framework for a Fixed-Wing Electric Vertical Takeoff and Landing (FW eVTOL) aircraft. The configuration consists of four rotors for vertical flight and one pusher rotor for cruise, integrated with six degrees of freedom (6DOF) flight dynamics. The simulation captures complex nonlinear flight behaviour through a comprehensive dynamic model while incorporating realistic sensor measurements from inertial measurement units (IMU). The sensor models provide realistic measurement data incorporating systematic errors and stochastic noise. The framework features translational and rotational dynamics, quaternion-based attitude representation, aerodynamic/thrust forces and moments modelling, atmospheric turbulence modelling using the Dryden turbulence model and high-fidelity actuator dynamics with physical constraints. Specifically, the propulsion system is modelled using full nonlinear Permanent Magnet Synchronous Motor (PMSM) equations. The simulation environment, developed in MATLAB/Simulink, employs fourth-order Runge-Kutta (RK4) numerical integration. The framework enables a comprehensive evaluation of flight control systems under realistic operational conditions.

3 Introduction

The FW eVTOL represents a transformative architecture in modern aviation, specifically for Urban Air Mobility (UAM) and cargo. Combining the vertical takeoff capabilities of a multi-rotor with the efficient cruise performance of a fixed-wing aircraft, it presents unique control challenges. Understanding and accurately modelling the complex flight dynamics of such hybrid aircraft, along with their associated systems, is paramount for advancing aerospace engineering and ensuring mission success in demanding operational environments.

Modern eVTOL aircraft rely heavily on integrated sensor systems to provide accurate state estimation for flight control, navigation, and mission systems. These sensors, including inertial measurement units (IMU), GNSS receivers, barometric altimeters and others must operate reliably across diverse flight conditions while maintaining precision in the presence of various error sources.

Flight dynamics simulation of hybrid aircraft presents unique challenges due to the inherently nonlinear, coupled, and multivariable nature of aerodynamic phenomena, particularly during the transition phase between hover and forward flight. The aircraft's capability to operate across a wide flight envelope necessitates sophisticated mathematical models that capture the intricate interplay between aerodynamic forces, inertial effects, control system dynamics, sensor measurements and actuator dynamics.

This report presents a comprehensive open-loop flight dynamics simulation framework for the FW-eVTOL aircraft, employing a nonlinear EoM mathematical model that encompasses complete six-degree-of-freedom dynamics. The simulation architecture integrates translational and rotational motion equations in the body frame, quaternion-based attitude representation to avoid singularities, detailed actuator dynamics with realistic physical constraints, comprehensive atmospheric turbulence modelling, and sophisticated sensor models incorporating both systematic bias and stochastic noise components.

The mathematical foundation is based on the first principles of Newtonian mechanics, incorporating the Newton-Euler equations of motion along with appropriate coordinate transformations between principal reference frames. The sensor models follow established s-domain transfer function representations with a dual-noise architecture that captures both low-frequency bias drift and high-frequency measurement noise characteristic of real sensor systems.

The simulation environment, developed within MATLAB's Simulink framework, provides a robust computational platform for numerical integration using different Numerical Solvers to solve a set of ordinary differential equations (ODEs), including the Euler Forward and fourth-order Runge-Kutta schemes. In this simulation, the 4th-order Runge-Kutta solver has been chosen for accuracy.

The model architecture is designed with software architecture's principles of modularity, scalability and maintainability in mind, facilitating future extensions for closed-loop flight control system development and validation. Instead of using built-in Simulink blocks to model the 6-DOF nonlinear dynamics, a user-defined function (MATLAB Function Block) was used to provide design freedom for implementing various design concepts. Figure 1 illustrates the input-output structure of the developed simulation framework, demonstrating the interconnected nature of the various subsystems.

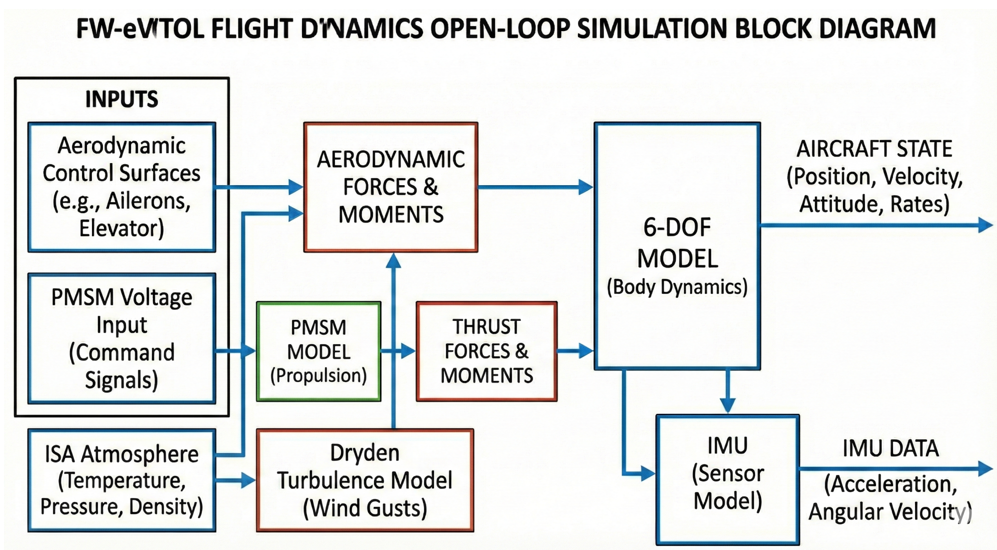

Figure 1: FW-eVTOL Flight Dynamics Simulation Model Architecture

3.1 Equations of Motion Block

The equations of motion governing aircraft dynamics are formulated through the application of Newton-Euler principles. In this specific formulation, we utilise the Body Frame for defining forces and moments, and the Earth-Centred Earth-Fixed (ECEF) frame for position integration.

- **Inertial Frame (North-East-Down, NED):** The inertial reference frame constitutes an Earth-fixed coordinate system wherein the positive X_e axis is oriented northward, the positive Y_e axis eastward, and the positive Z_e axis directs downward in alignment with the local gravity vector.

- **Body-Fixed Frame:** The body axis system is rigidly affixed to the aircraft's centre of gravity (CG). The positive X_b axis extends forward through the aircraft nose, the positive Y_b axis projects laterally toward the starboard wingtip, and the positive Z_b axis points downward.

These coordinate systems, along with their associated transformation matrices, form the mathematical foundation for expressing the complete six-degree-of-freedom motion of the aircraft, encompassing both translational and rotational dynamics in three-dimensional space.

3.2 Translational Dynamics (Body Frame)

The translational equations of motion are expressed in the body frame. We directly integrate the body velocities u, v, w:

$$\dot{u} = rv - qw - g\sin\theta + \frac{F_{X_{aero}} + F_{X_{prop}}}{m} \tag{1}$$

$$\dot{v} = -ru + pw + g\cos\theta\sin\phi + \frac{F_{Y_{aero}} + F_{Y_{prop}}}{m} \tag{2}$$

$$\dot{w} = qu - pv + g\cos\theta\cos\phi + \frac{F_{Z_{aero}} + F_{Z_{prop}}}{m} \tag{3}$$

Here, F_{aero} represents aerodynamic forces (Lift, Drag, Side force projected to body axes) and F_{prop} represents forces generated by the 5 rotors.

3.3 Rotational Dynamics

$$\dot{p} = \frac{I_{zz}(L_a + L_p) + I_{xz}(N_a + N_p)}{\Delta} + \frac{\big((I_{xx} - I_{yy})I_{xz} + I_{xz}I_{zz}\big)pq}{\Delta} + \frac{(I_{yy}I_{zz} - I_{zz}^2 - I_{xz}^2)qr}{\Delta} \tag{4}$$

$$\dot{q} = \frac{M_a + M_p}{I_{yy}} - \frac{(I_{xx} - I_{zz})pr}{I_{yy}} - \frac{I_{xz}(p^2 - r^2)}{I_{yy}} \tag{5}$$

$$\dot{r} = \frac{I_{xz}(L_a + L_p) + I_{xx}(N_a + N_p)}{\Delta} + \frac{(I_{xx}^2 - I_{xx}I_{yy} + I_{xz}^2)pq}{\Delta} - \frac{I_{xz}(I_{xx} - I_{yy} + I_{zz})qr}{\Delta} \tag{6}$$

where $\Delta = I_{xx} \cdot I_{zz} - I_{xz}^2$ and I_{xx},I_{yy},I_{zz} are the moments of inertia. L_a, M_a, N_a L_p, M_p, N_p are moments generated by the aerodynamic surfaces and propulsion system.

3.4 Quaternion Kinematics

$$\dot{\mathbf{q}} = \tfrac{1}{2}\,\mathbf{q} \otimes [0, p, q, r]^\top \tag{7}$$

3.5 NED Position Rates

The position is propagated in the NED frame using the body-frame velocity components transformed through the rotation matrix:

$$\begin{bmatrix} \dot{P}_X \\ \dot{P}_Y \\ \dot{P}_Z \end{bmatrix}_{NED} = R_b^{NED} \begin{bmatrix} u \\ v \\ w \end{bmatrix} \tag{8}$$

where R_b^{NED} is the rotation matrix from the body frame to the NED frame, derived from the quaternion representation $\mathbf{q} = [q_0, q_1, q_2, q_3]^T$:

$$R_b^{NED} = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix} \tag{9}$$

where q_0 is the scalar part and q_1, q_2, q_3 are the vector components of the unit quaternion.

4 Actuator Models

The actuators are critical components of the flight control system, providing the motive power needed to move the flight control surfaces. For the FW- eVTOL, this includes conventional aerodynamic surfaces and an electric propulsion system.

4.1 Control Surface Actuators

The primary flight control surfaces, namely aileron (δ_a), elevator (δ_e), and rudder (δ_r), are shown in Figure 2,3. These conventional control surfaces provide aerodynamic control primarily during forward flight.

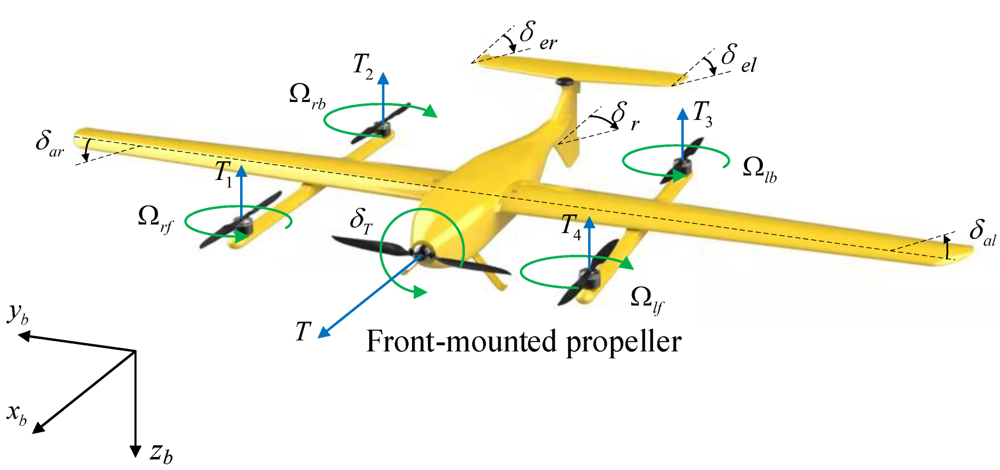

Figure 2: FW eVTOL aerodynamic control surfaces

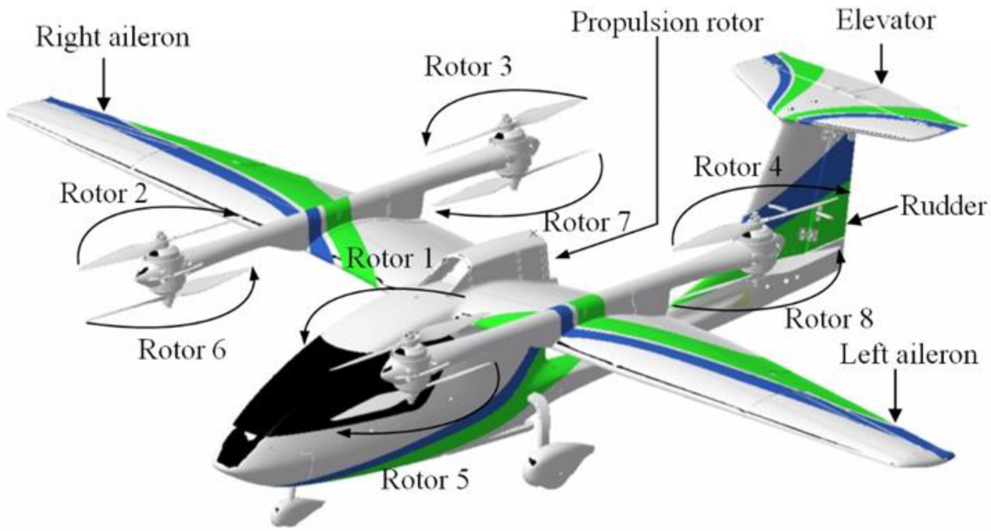

Figure 3: FW eVTOL aerodynamic control surfaces

The actuator mathematical models for the control surfaces are represented as second-order transfer functions with pure time delay

$$\frac{\delta_i(s)}{\delta_{i,\text{com}}(s)} = K\, e^{-s\tau}\, \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{10}$$

where:

- K: steady-state gain,
- ω_n: natural frequency [rad/s],
- ζ: damping ratio.
- τ: time delay.
- $i \in \{a, e, r\}$ (aileron, elevator, rudder)

For implementation, the delay term $e^{-s\tau}$ can be approximated using 1-order Padé expansions as:

$$e^{-s\tau} \approx \frac{1 - \frac{\tau}{2}s}{1 + \frac{\tau}{2}s} \tag{11}$$

for $K = 1$, $\omega_n = 75$ rad/s, $\zeta = 0.7$ and $\tau = 0.008\,$s. We will have

$$\frac{\delta_i(s)}{\delta_{i,\text{com}}(s)} = \frac{75^2\left(1 - 0.004\, s\right)}{\left(1 + 0.004\, s\right)\left(s^2 + 105\, s + 75^2\right)} \tag{12}$$

4.2 Electric Propulsion Actuators (Nonlinear PMSM Model)

The 5 rotors (4 lift, 1 cruise) are driven by Permanent Magnet Synchronous Motors (PMSM). A full nonlinear model is implemented for each motor in the ABC reference frame.

4.2.1 Electrical Dynamics

The stator current dynamics are modeled in the natural ABC frame with position-dependent inductances:

$$\frac{di_a}{dt} = \frac{1}{L_{aa}} (v_a - R_s i_a - e_a) \tag{13}$$

$$\frac{di_b}{dt} = \frac{1}{L_{bb}} (v_b - R_s i_b - e_b) \tag{14}$$

$$\frac{di_c}{dt} = \frac{1}{L_{cc}} (v_c - R_s i_c - e_c) \tag{15}$$

where the self-inductances vary with the electrical rotor position θ_e according to:

$$L_{aa} = L_s + L_m \cos(2\theta_e) \tag{16}$$

$$L_{bb} = L_s + L_m \cos(2\theta_e - 2\pi/3) \tag{17}$$

$$L_{cc} = L_s + L_m \cos(2\theta_e - 4\pi/3) \tag{18}$$

with $L_s = (L_d + L_q)/2$ and $L_m = (L_d - L_q)/2$ representing the average and differential inductances, respectively.

The back-EMF terms are given by:

$$e_a = -\omega_e \lambda_{pm} \sin(\theta_e) \tag{19}$$

$$e_b = -\omega_e \lambda_{pm} \sin(\theta_e - 2\pi/3) \tag{20}$$

$$e_c = -\omega_e \lambda_{pm} \sin(\theta_e - 4\pi/3) \tag{21}$$

where:

- v_a, v_b, v_c: Three-phase stator voltages (control inputs)
- i_a, i_b, i_c: Three-phase stator currents
- R_s: Stator resistance
- λ_{pm}: Permanent magnet flux linkage
- $\omega_e = p\omega_m$: Electrical angular velocity (p is number of pole pairs)
- θ_e: Electrical rotor position

4.2.2 Mechanical Dynamics

The rotor speed dynamics are governed by:

$$\frac{d\omega_m}{dt} = \alpha = \frac{1}{J} (T_e - T_L - B\omega_m) \tag{22}$$

The electromagnetic torque T_e is calculated from the dq-frame currents obtained via the Park transformation:

$$T_e = \frac{3}{2} p \left(\lambda_{pm} i_q + (L_d - L_q) i_d i_q\right) \tag{23}$$

The inductances L_d and L_q are functions of the dq-frame currents.
The load torque T_L represents the aerodynamic drag of the propeller.
The electrical angle evolves according to:

$$\frac{d\theta_e}{dt} = \omega_e = p\omega_m \tag{24}$$

Each of the 5 motors is modelled independently using these equations with protection limits.

4.3 Control Input Constraints

The control input vector is defined as:

$$\mathbf{U} = [\delta_a, \delta_e, \delta_r, \omega_p, \omega_{r,1..4}]^T \tag{25}$$

where δ_a, δ_e, and δ_r represent the deflection commands for the aerodynamic surfaces, and ω_p and $\omega_{r,1..4}$ denote the angular velocity commands for the cruise propeller and lift rotors, respectively.

The operational limits for the aerodynamic surfaces are:

$$-25^\circ \le \delta_e \le 25^\circ \quad \text{(elevator)} \tag{26}$$
$$-25^\circ \le \delta_a \le 25^\circ \quad \text{(aileron)} \tag{27}$$
$$-25^\circ \le \delta_r \le 25^\circ \quad \text{(rudder)} \tag{28}$$

The complete set of actuator system parameters is presented in Table 1, which specifies the kinematic constraints, bandwidth characteristics, and computational latencies.

Table 1: Aerodynamic actuator model parameters

Control Surface	**Position Limit**	**Rate Limit**	ω_n **(rad/s)**	ζ	τ **(ms)**
$\delta_{a,e,r}$	±25	±80/s	75	0.6	8

Figure 4 presents the Simulink implementation of the aerodynamic actuation system, illustrating the integration of second-order actuator dynamics with rate and position saturation blocks.

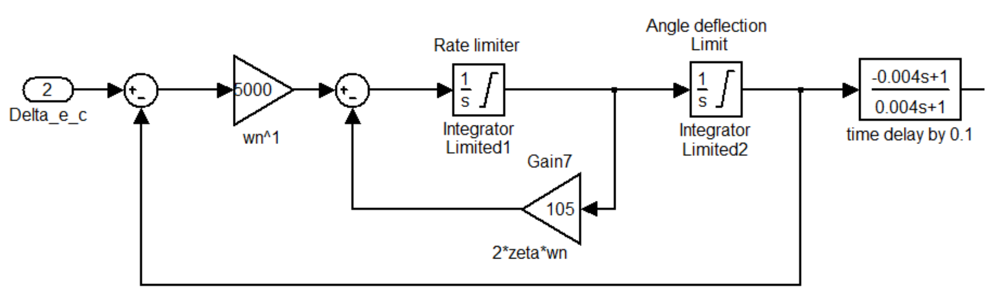

Figure 4: Simulink block implementation of the aerodynamic actuation system

The PMSM motors are subject to the following electrical and mechanical constraints:

$$|v_{\text{phase}}| \leq v_{\max} = 300\,\text{V} \quad \text{(phase voltage limit)} \tag{29}$$

$$|i_{\text{phase}}| \leq i_{\max} = 20\,\text{A} \quad \text{(phase current limit)} \tag{30}$$

$$\omega_m \in [-52.36, 52.36]\,\text{rad/s} \quad \text{(angular velocity limit)} \tag{31}$$

$$\alpha \in [-2000, 2000]\,\text{rad/s}^2 \quad \text{(angular acceleration limit)} \tag{32}$$

These saturation limits are rigorously enforced throughout the simulation to ensure physical realizability of control commands and maintain actuator responses within certified operational envelopes. The hybrid actuation architecture, combining conventional aerodynamic control surfaces with distributed electric propulsion, endows the eVTOL with enhanced control authority and maneuverability across all flight regimes, from hover through transition to cruise flight.

This actuator modelling framework provides a high-fidelity representation of modern flight actuation systems, which permits straightforward adaptation to alternative hardware specifications.

5 Atmospheric Model (ISA)

As an aircraft operates at various altitudes, the aerodynamic coefficients need to account for changes in density as represented by a standard atmosphere model. The International Standard Atmosphere (ISA) model used in this simulation provides standard atmospheric values for air temperature T, pressure P, density ρ, speed of sound a, and dynamic pressure q_{bar}, as a function of the altitude h and true airspeed V_T.

5.1 Reference Constants

The ISA model uses the following sea-level reference conditions:

$$T_0 = 288.15 \text{ K} \quad \text{(sea-level temperature)} \tag{33}$$

$$P_0 = 101325 \text{ Pa} \quad \text{(sea-level pressure)} \tag{34}$$

$$R = 287.05 \text{ J/(kg·K)} \quad \text{(specific gas constant for air)} \tag{35}$$

$$g = 9.80665 \text{ m/s}^2 \quad \text{(standard gravity)} \tag{36}$$

$$\gamma = 1.4 \quad \text{(heat capacity ratio)} \tag{37}$$

5.2 Temperature and Pressure Calculation

The atmospheric properties are computed using the hydrostatic equation and the ideal gas law based on altitude layers.

Troposphere ($h \leq h_1 = 11000$ m):

$$T = T_0 + L_1 h \tag{38}$$

$$P = P_0 \left(\frac{T}{T_0} \right)^{-\frac{g}{RL_1}} \tag{39}$$

where $L_1 = -0.0065$ K/m is the temperature lapse rate in the troposphere.
Above Troposphere ($h > 11000$ m):

For altitudes above 11 km, an isothermal approximation is used:

$$T_1 = T_0 + L_1 h_1 \tag{40}$$

$$P_1 = P_0 \left(\frac{T_1}{T_0}\right)^{-\frac{g}{RL_1}} \tag{41}$$

$$T = T_1 \tag{42}$$

$$P = P_1 \exp\left(-\frac{g(h - h_1)}{RT_1}\right) \tag{43}$$

5.3 Derived Quantities

Once temperature and pressure are determined, the remaining atmospheric properties are calculated using fundamental thermodynamic relationships:
Air density:

$$\rho = \frac{P}{RT} \tag{44}$$

Dynamic pressure:

$$q_{\text{bar}} = \frac{1}{2}\rho V_T^2 \tag{45}$$

Speed of sound:

$$a = \sqrt{\gamma RT} \tag{46}$$

Mach number:

$$M = \frac{V_T}{a} \tag{47}$$

6 Atmospheric Turbulence Modeling

Real-world flight operations are significantly influenced by atmospheric turbulence, which originates from atmospheric pressure instabilities and non-uniform temperature distributions, introducing stochastic disturbances that affect aircraft stability, handling qualities, and structural loads. The incorporation of accurate turbulence models is essential for comprehensive flight simulation and robust control system validation under realistic atmospheric conditions.

The inherently stochastic nature of atmospheric turbulence cannot be modelled exactly, necessitating statistical approaches that capture the essential spectral characteristics of turbulent velocity fields. Several well-established turbulence models have been used in the literature to implement turbulence, among which the Dryden and von Kármán spectral models have achieved widespread acceptance due to their physical realism, mathematical simplicity, and computational efficiency.

6.1 Dryden Turbulence Model

The Dryden turbulence model is widely adopted in aerospace engineering due to its mathematical tractability and good correlation with measured atmospheric turbulence spectra. The model generates correlated turbulence velocity components through shaping filters that transform white noise inputs into realistic atmospheric disturbances with appropriate spectral characteristics.

The Dryden model defines three turbulence velocity components corresponding to longitudinal (u_{turb}), lateral (v_{turb}), and vertical (w_{turb}) atmospheric disturbances. These components are generated using the following continuous-domain transfer functions:

$$G_{ug}(s) = \sigma_u \sqrt{\frac{2L_u}{\pi V}} \cdot \frac{1}{1 + \frac{L_u}{V} s} \tag{48}$$

$$G_{vg}(s) = \sigma_v \sqrt{\frac{2L_v}{\pi V}} \cdot \frac{1 + \frac{2\sqrt{3}L_v}{V} s}{\left(1 + \frac{2L_v}{V} s\right)^2} \tag{49}$$

$$G_{wg}(s) = \sigma_w \sqrt{\frac{2L_w}{\pi V}} \cdot \frac{1 + \frac{2\sqrt{3}L_w}{V} s}{\left(1 + \frac{2L_w}{V} s\right)^2} \tag{50}$$

where σ_u, σ_v, σ_w are the turbulence intensity parameters, L_u, L_v, L_w are the turbulence length scales, and V is the aircraft airspeed. The longitudinal turbulence exhibits first-order dynamics, while the lateral and vertical components follow second-order characteristics to capture the more complex cross-coupling effects observed in atmospheric measurements.

6.2 Discrete-Time Implementation

The continuous-domain Dryden filters are converted to discrete-time equivalents using the bilinear transformation (Tustin method) to enable numerical integration within the simulation framework. This transformation preserves the essential spectral characteristics while ensuring numerical stability at the chosen integration timestep.

The bilinear transformation $s = \frac{2}{dt}\frac{z-1}{z+1}$ is applied to each transfer function, resulting in discrete-time difference equations suitable for real-time implementation. The transformed filters maintain appropriate filter memory states to ensure continuity and realistic temporal correlation in the generated turbulence sequences.

6.3 Turbulence-Induced Aerodynamic Perturbations

Atmospheric turbulence modifies the relative wind velocity vector experienced by the aircraft. The turbulence components in the body-fixed reference frame are denoted as u_{turb}, v_{turb}, and w_{turb}, representing longitudinal, lateral, and vertical wind disturbances, respectively. These components are generated through spectral filtering of white noise to produce realistic atmospheric disturbances with appropriate temporal and spatial correlation characteristics.

6.3.1 Linear Velocity Perturbations

The relative wind velocity experienced by the aircraft is computed by subtracting the turbulence velocity components from the aircraft's body-frame velocity components:

$$u_{rel} = u - u_{turb} \tag{51}$$

$$v_{rel} = v - v_{turb} \tag{52}$$

$$w_{rel} = w - w_{turb} \tag{53}$$

The magnitude of the relative velocity and the aerodynamic angles are computed from the perturbed velocity components:

$$V_{rel} = \sqrt{u_{rel}^2 + v_{rel}^2 + w_{rel}^2} \tag{54}$$

$$\alpha = \arctan\left(\frac{w_{rel}}{u_{rel}}\right) \tag{55}$$

$$\beta = \arcsin\left(\frac{v_{rel}}{V_{rel}}\right) \tag{56}$$

where α is the angle of attack and β is the sideslip angle, both modified by the presence of atmospheric turbulence.

6.3.2 Angular Rate Perturbations

In addition to linear velocity turbulence, atmospheric disturbances induce rotational perturbations on the aircraft. These angular rate disturbances arise from spatial gradients in the turbulent velocity field and can be approximated using the following relations based on MIL-STD-1797A:

$$p_{gust} = \frac{\pi}{4b}\left(w_{turb} - v_{turb}\right) \tag{57}$$

$$q_{gust} = \frac{\pi}{3b} w_{turb} \tag{58}$$

$$r_{gust} = \frac{\pi}{4b}\left(v_{turb} + w_{turb}\right) \tag{59}$$

where b is the aircraft wingspan and p_{gust}, q_{gust}, r_{gust} represent roll, pitch, and yaw rate disturbances, respectively. These angular perturbations modify the effective body angular rates experienced by the aerodynamic surfaces, thereby affecting the damping and rate-dependent moment contributions.

The relative angular rates become:

$$p_{rel} = p - p_{gust} \tag{60}$$

$$q_{rel} = q - q_{gust} \tag{61}$$

$$r_{rel} = r - r_{gust} \tag{62}$$

7 Aerodynamic Force and Moment

7.1 Aerodynamic Force

The aerodynamic forces are computed using the disturbed values V_{rel}, α, and β in place of the undisturbed values.

The aerodynamic forces are initially computed in the wind-axis, where:

$$L = q_{bar}\, S\, C_L \quad \text{(Lift, perpendicular to } V_{rel}) \tag{63}$$

$$D = q_{bar}\, S\, C_D \quad \text{(Drag, parallel to } V_{rel}) \tag{64}$$

$$Y = q_{bar}\, S\, C_Y \quad \text{(Side force)} \tag{65}$$

where S is the reference wing area, and q_{bar} is the dynamic pressure calculated using the relative velocity:

$$q_{bar} = \frac{1}{2}\rho V_{rel}^2 \tag{66}$$

C_L,C_D, C_Y are the aerodynamic force coefficients which are evaluated using perturbed aerodynamic angles.

An example can be represented as:

$$C_L = C_{L_0} + C_{L_\alpha}\alpha + C_{L_q}\frac{q_{rel}\bar{c}}{2V_{rel}} + C_{L_{\delta_e}}\delta_e \tag{67}$$

$$C_D = C_{D_0} + C_{D_\alpha}\alpha^2 + C_{D_{\delta_e}}|\delta_e| \tag{68}$$

$$C_Y = C_{Y_\beta}\beta + C_{Y_p}\frac{p_{rel}b}{2V_{rel}} + C_{Y_r}\frac{r_{rel}b}{2V_{rel}} + C_{Y_{\delta_a}}\delta_a + C_{Y_{\delta_r}}\delta_r \tag{69}$$

7.2 Aerodynamic Moment

The aerodynamic moments about the aircraft center of gravity are similarly computed using the relative velocity and angular rates.

The aerodynamic moments about the body-fixed axes are:

$$L_{aero} = q_{bar}\, S\, b\, C_\ell \tag{70}$$

$$M_{aero} = q_{bar}\, S\, \bar{c}\, C_m \tag{71}$$

$$N_{aero} = q_{bar}\, S\, b\, C_n \tag{72}$$

where b is the wingspan and $\bar{c}$ is the mean aerodynamic chord.

C_l,C_m, C_n are the moment coefficients represented for example as:

$$C_\ell = C_{\ell_\beta}\beta + C_{\ell_p}\frac{p_{rel}b}{2V_{rel}} + C_{\ell_r}\frac{r_{rel}b}{2V_{rel}} + C_{\ell_{\delta_a}}\delta_a + C_{\ell_{\delta_r}}\delta_r \tag{73}$$

$$C_m = C_{m_0} + C_{m_\alpha}\alpha + C_{m_q}\frac{q_{rel}\bar{c}}{2V_{rel}} + C_{m_{\delta_e}}\delta_e \tag{74}$$

$$C_n = C_{n_\beta}\beta + C_{n_p}\frac{p_{rel}b}{2V_{rel}} + C_{n_r}\frac{r_{rel}b}{2V_{rel}} + C_{n_{\delta_a}}\delta_a + C_{n_{\delta_r}}\delta_r \tag{75}$$

C_{L_α}, ,C_L
,C_{Y_β},C_{Y_δ}, etc., are the stability derivatives, which need to be adjusted for the specific airframe geometry of the FW-eVTOL. Note: for this simulation, random values have been chosen for testing purposes.

8 Motor Thrust Forces and Moments

The FW-eVTOL configuration features four vertical takeoff and landing (VTOL) rotors for lift generation and one cruise propeller in a pusher configuration for forward propulsion.

8.1 Thrust Generation

The thrust produced by each rotor i is proportional to the square of its angular velocity ω_i:

$$T_i = K_T\omega_i^2 \tag{76}$$

where K_T is the thrust coefficient. For the VTOL rotors, $i \in \{1, 2, 3, 4\}$, and the cruise propeller is denoted by subscript p.

8.2 Force Components

The propulsion forces are computed in the body frame, where the individual rotor thrust vectors are defined relative to their respective mounting positions. The VTOL rotor forces are expressed as:

$$\mathbf{F}_{vtol,i} = \begin{bmatrix} 0 \\ 0 \\ -T_i \end{bmatrix}, \quad i = 1, 2, 3, 4 \tag{77}$$

noting that the negative sign accounts for the body frame convention where positive Z points downward, while rotor thrust acts upward.

The cruise propeller generates longitudinal thrust:

$$\mathbf{F}_{prop} = \begin{bmatrix} T_p \\ 0 \\ 0 \end{bmatrix} \tag{78}$$

The total thrust force in the body frame is:

$$\mathbf{F}_{thrust} = \mathbf{F}_{prop} + \sum_{i=1}^{4}\mathbf{F}_{vtol,i} = \begin{bmatrix} T_p \\ 0 \\ -\sum_{i=1}^{4} T_i \end{bmatrix} \tag{79}$$

8.3 Moment Components

The moments generated by the propulsion system are computed using the cross product of the position vectors from the center of gravity to each rotor and their respective thrust forces. For the VTOL rotors positioned at $\mathbf{r}_{vtol,i} = [x_i, y_i, z_i]^T$:

$$\mathbf{M}_{vtol,i} = \mathbf{r}_{vtol,i} \times \mathbf{F}_{vtol,i} \tag{80}$$

The contribution from the cruise propeller, positioned at $\mathbf{r}_p$, is:

$$\mathbf{M}_{prop} = \mathbf{r}_p \times \mathbf{F}_{prop} \tag{81}$$

The total moment about the center of gravity is:

$$\mathbf{M}_{thrust} = \mathbf{M}_{prop} + \sum_{i=1}^{4} \mathbf{M}_{vtol,i} \tag{82}$$

9 Sensors Model

eVTOL aircrafts are critically dependent on integrated sensor systems to provide accurate state estimation for flight control, navigation, and mission systems. The eVTOL simulation framework incorporates a comprehensive multisensor model that includes inertial measurement units (IMU) with realistic error characteristics and dynamic responses.

10 Sensor Model Architecture

The multi-sensor suite is represented by a unified, five-stage processing chain that captures the principal characteristics of practical sensing systems:

1. **Dynamic filtering:** sensor dynamics described by a transfer function.
2. **Scale transformation:** application of sensor-specific scale (gain) factors.
3. **Bias injection:** addition of slowly varying systematic offsets.
4. **Noise corruption:** addition of measurement noise (high-frequency).
5. **Time delay:** latency due to digital processing and communication.

A general time-domain model for a single measurement channel is

$$y(t) = K\,\mathcal{H}\{u(t-T)\} + b(t-T) + n(t-T) \tag{83}$$

where $\mathcal{H}$ represents the sensor transfer function, K is the scale factor, $u(t)$ the true input, T the pure time delay, $b(t)$ the bias process, and $n(t)$ the additive measurement noise.

In the Laplace domain:

$$Y(s) = K\,H(s)\,e^{-sT}U(s) + e^{-sT}\big(B(s) + N(s)\big) \tag{84}$$

i.e. the delay operator e^{-sT} applies to the complete processed signal.

10.1 Bias dynamics

Sensor bias denotes a slowly varying systematic error caused by temperature, ageing, calibration drift and similar effects. A common and practical model is the first-order Gauss-Markov process

$$\dot{b}(t) = -\frac{1}{\tau}\, b(t) + w(t) \tag{85}$$

where $\tau > 0$ is the bias correlation time and $w(t)$ is zero-mean white driving noise with variance σ_w^2. The corresponding transfer function from the driving noise to the bias is

$$\frac{B(s)}{W(s)} = \frac{\tau}{\tau s + 1}. \tag{86}$$

where the bias variance is

$$\sigma_b^2 = \frac{\sigma_w^2\,\tau}{2} \qquad \Longrightarrow \qquad \sigma_w^2 = \frac{2\sigma_b^2}{\tau} \tag{87}$$

10.2 Measurement noise

High-frequency measurement disturbances (e.g. ADC quantisation, electronic noise, small-scale turbulence) are modelled as additive, zero-mean Gaussian white noise:

$$n(t) \sim \mathcal{N}(0, \sigma_n^2), \tag{88}$$

where σ_n^2 denotes the noise variance.

10.3 Inertial Measurement Unit (IMU)

The IMU provides three-axis acceleration and angular rate measurements through accelerometer and gyroscope sensor clusters. These sensors are fundamental to the aircraft's inertial navigation system and flight control loops. The sensor families follow the general architecture above.

10.3.1 Accelerometer

The accelerometer measurement vector is modelled as

$$\mathbf{y}_{\mathrm{acc}}(t) = K_{\mathrm{acc}}\, \mathcal{H}_{\mathrm{acc}}\{\mathbf{f}_{\mathrm{true}}(t-T)\} + \mathbf{b}_{\mathrm{acc}}(t-T) + \mathbf{n}_{\mathrm{acc}}(t-T), \tag{89}$$

with a second-order dynamics approximation

$$H_{\mathrm{acc}}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \tag{90}$$

and bias dynamics governed by the Gauss-Markov model in which each axis follows the same form as (3). Typical parameter values (representative) are:

- natural frequency: $\omega_n = 100$ rad/s ($\approx$15.9 Hz)

- damping ratio: $\zeta = 0.7$ (underdamped)
- bias correlation time: $\tau_{\text{acc}} = 300$ s
- bias std: $\sigma_{b,\text{acc}} = 0.0000485$ m/s^2
- noise variance: $\sigma^2_{n,\text{gyro}} = 0.00276$ rad/s/$\sqrt{\text{Hz}}$.

10.3.2 Gyroscope

The gyroscope measurement is modelled as

$$\boldsymbol{\omega}_{\text{meas}}(t) = K_{\text{gyro}}\,\mathcal{H}_{\text{gyro}}\{\boldsymbol{\omega}_{\text{true}}(t-T)\} + \boldsymbol{b}_{\text{gyro}}(t-T) + \boldsymbol{n}_{\text{gyro}}(t-T) \tag{91}$$

with a first-order dynamics approximation

$$H_{\text{gyro}}(s) = \frac{1}{\tau_{\text{gyro,d}}s + 1} \tag{92}$$

and Gauss–Markov bias dynamics per axis. Representative parameters are:

- scale factor: $K_{\text{gyro}} = 1.0 \pm 0.002$ (0.2% error)
- bias correlation time: $\tau_{\text{gyro,b}} = 300$ s
- bias std: $\sigma_{b,\text{acc}} = 0.0000485$ m/s^2
- noise variance: $\sigma^2_{n,\text{gyro}} = 0.00276$ rad/s/$\sqrt{\text{Hz}}$.

Note: the above bias and noise values are derived from bias instability of 10°/h: $\mathbf{b} =$
4.85×10^{-5} rad/s

and angular random walk (ARW) of 0.005°/s/$\sqrt{\text{Hz}}$ at sampling frequency $f_s = 1000$ Hz

Figure 5 illustrates the Simulink implementation of this architecture for the gyroscope, demonstrating the cascaded processing stages from true altitude input to corrupted measurement output.

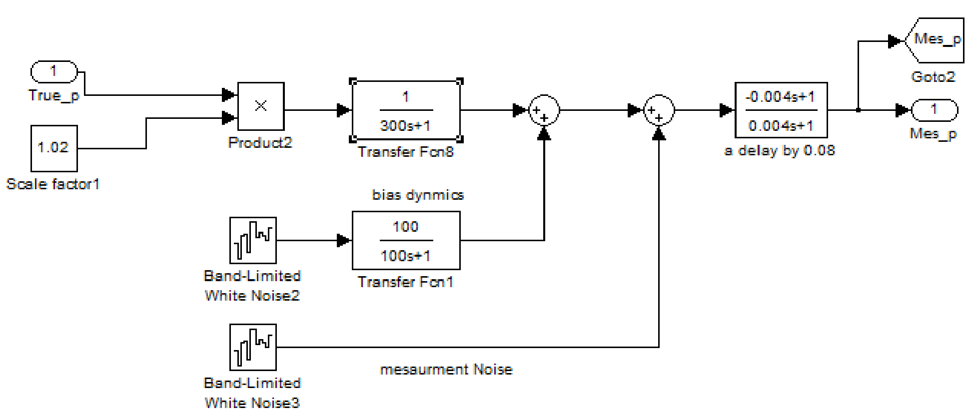

Figure 5: Simulink block implementation of the gyroscope sensor model

11 Simulation Setup

The simulation has been implemented in MATLAB R2011a with the following initial conditions and parametres:

- **Numerical Integration:** Fourth-order Runge-Kutta (RK4) method with time step $\Delta t = 0.001\,\text{s}$ and total simulation time $T = 50\,\text{s}$.
- **Initial Conditions:**
  - Position: $\mathbf{r}_0 = [0, 0, -1300]^\top\,\text{m}$
  - Velocity: $u = 25\,\text{m/s}$, $v = w = 0\,\text{m/s}$
  - Quaternion: $\mathbf{q}_0 = [0, 0.4, 0.6, 0]^\top$
  - Angular rates: $p = q = r = 0\,\text{rad/s}$
- **Aircraft Parameters:**
  - Mass: $m = 650\,\text{kg}$
  - Moments of inertia: $I_{xx} = 450\,\text{kg·m}^2$, $I_{yy} = 650\,\text{kg·m}^2$, $I_{zz} = 850\,\text{kg·m}^2$
  - Gravitational acceleration: $g = 9.80\,\text{m/s}^2$
  - Wing area: $S = 8.5\,\text{m}^2$
  - Wing span: $b = 11\,\text{m}$
  - Mean aerodynamic chord: $c = 0.77\,\text{m}$
- **Propulsion System Configuration:**
  - VTOL rotor positions (quad configuration):

$$
\begin{aligned}
\mathbf{r}_{\text{vtol},1} &= [1.2, 1.2, -0.1]^\top\,\text{m} \quad \text{(front-right)}\\
\mathbf{r}_{\text{vtol},2} &= [1.2, -1.2, -0.1]^\top\,\text{m} \quad \text{(front-left)}\\
\mathbf{r}_{\text{vtol},3} &= [-1.2, -1.2, -0.1]^\top\,\text{m} \quad \text{(rear-left)}\\
\mathbf{r}_{\text{vtol},4} &= [-1.2, 1.2, -0.1]^\top\,\text{m} \quad \text{(rear-right)}
\end{aligned}
$$

  - Pusher propeller position: $\mathbf{r}_\text{p} = [-2.5, 0, 0]^\top\,\text{m}$
  - VTOL rotor thrust coefficients: $k_{T,i} = 0.000185\,\text{N}$ for $i = 1, \ldots, 4$
  - Cruise propeller thrust coefficient: $k_{T,\text{p}} = 0.00045\,\text{N}/$
- **PMSM Electrical Parameters:**
  - Stator resistance: $R_s = 0.0285\,\Omega$
  - d-axis inductance: $L_d = 0.032\,\text{H}$
  - q-axis inductance: $L_q = 0.0032\,\text{H}$
  - Permanent magnet flux linkage: $\lambda_{\text{pm}} = 0.078\,\text{Wb}$

- Rotor inertia: $J = 0.00025\,\text{kg}\cdot\text{m}^2$
- Viscous friction coefficient: $B = 0.0001\,\text{N}\cdot\text{m}\cdot\text{s}$
- Number of pole pairs: $p = 4$
- Angular velocity limits: $\omega \in [-52.36, 52.36]\,\text{rad/s}$
- Angular acceleration limits: $\alpha \in [-2000, 2000]\,\text{rad/s}^2$
- Current limit: $i_{\max} = 20\,\text{A}$
- Voltage limit: $v_{\max} = 300\,\text{V}$

- **Atmospheric Disturbances:** Dryden turbulence model with light intensity, turbulence filters initialised at zero states.

12 Results

The following results showcase some of the open-loop flight dynamic response.

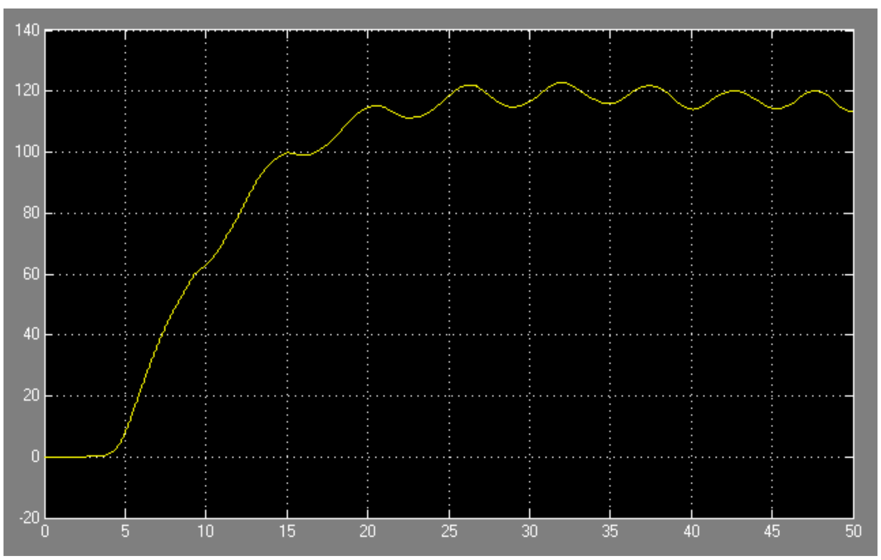

Figure 6: x Position.

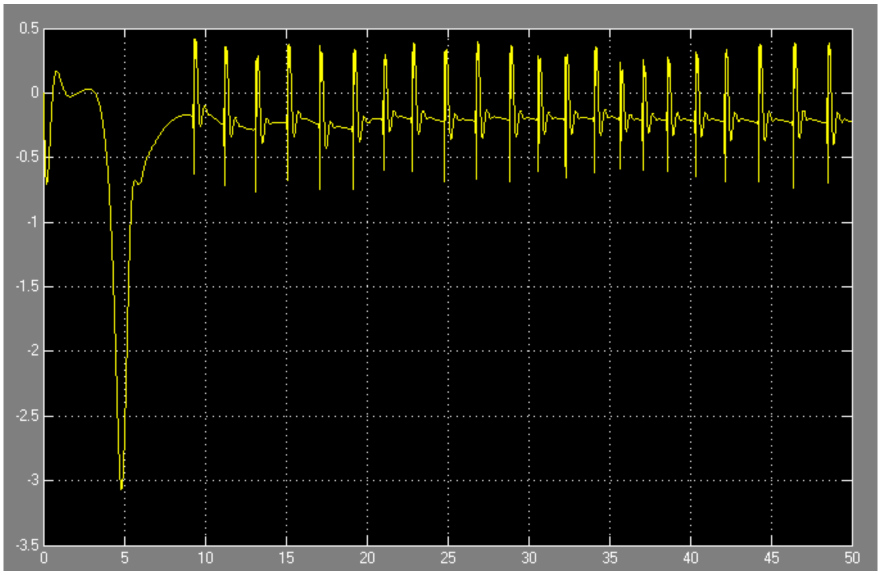

Figure 7: Yaw angular rate r.

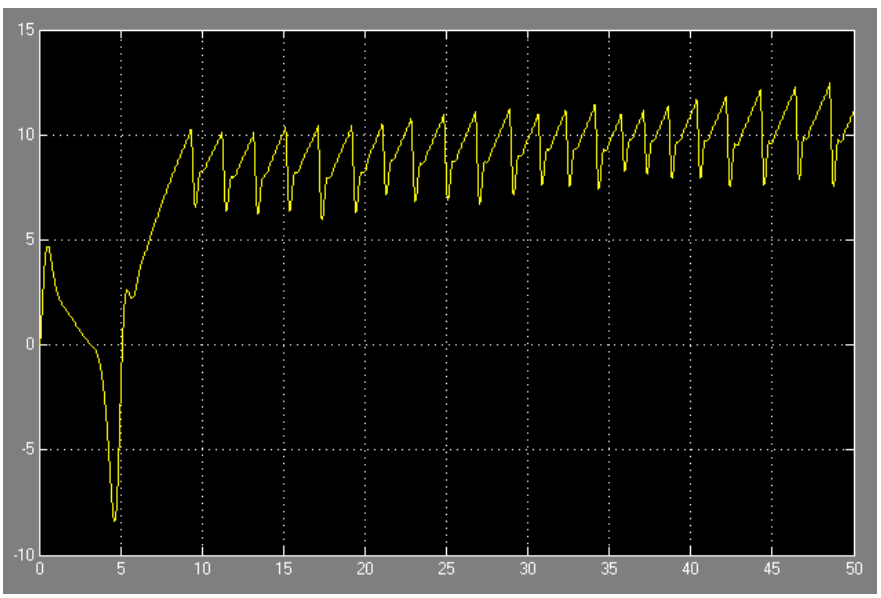

Figure 8: Body linear vertical velocity w.

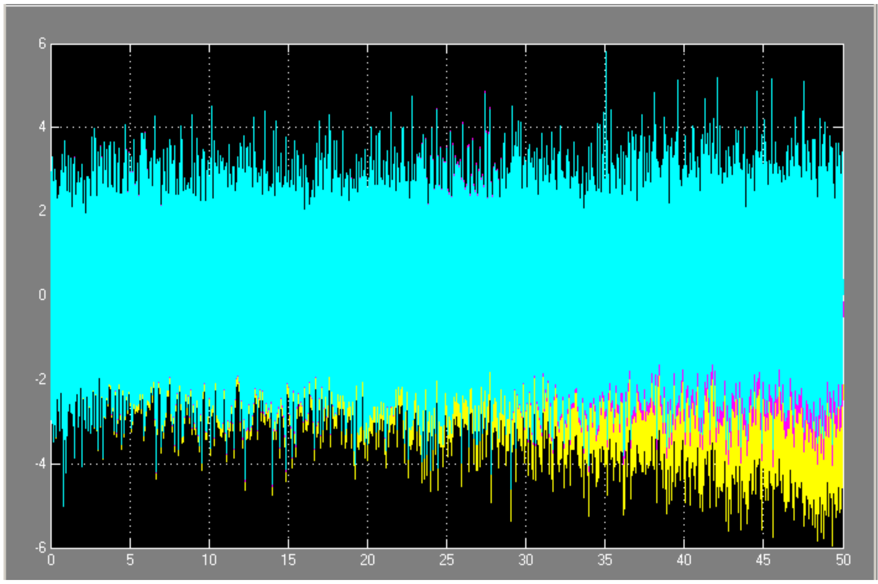

Figure 9: Gyroscope sensor's output

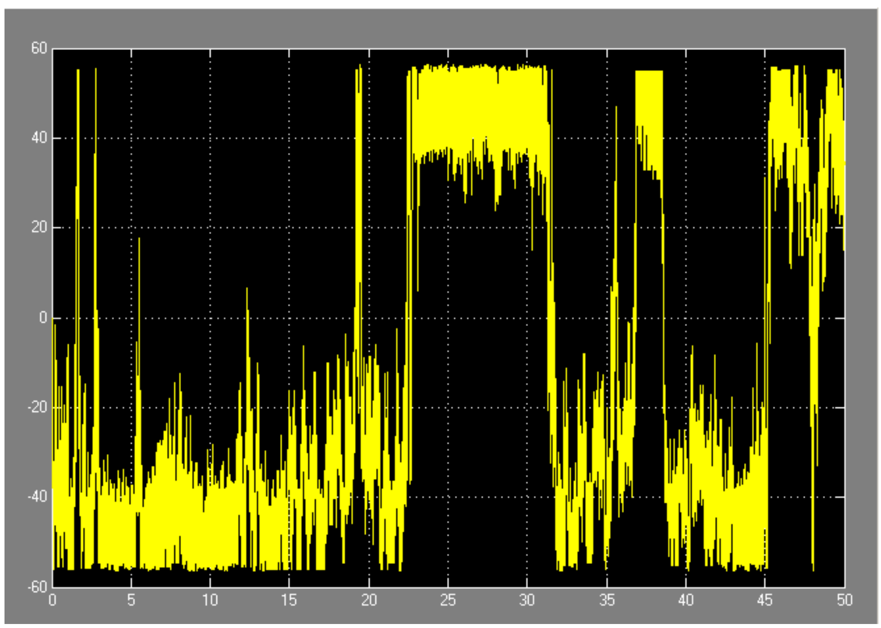

Figure 10: Pusher PMSM mechanical rotor speed output

13 Conclusion

This work presents a comprehensive high-fidelity simulation framework for the eVTOL aircraft, successfully integrating six-degree-of-freedom nonlinear flight dynamics with realistic sensor models and atmospheric disturbances. The implementation demonstrates several key achievements that establish a robust foundation for advanced flight control research.

The numerical architecture, employing fourth-order Runge-Kutta integration with 1 ms timestep, maintained stable behaviour throughout extended simulation periods without numerical instabilities. The adoption of quaternion-based attitude representation effectively eliminated the gimbal lock singularities inherent to Euler angle formulations, ensuring stable flight across the entire attitude envelope, including extreme pitch conditions near ± 90.

The actuator subsystem accurately captures the physical constraints and dynamic characteristics of modern eVTOL flight control systems. Position and rate limits, combined with realistic time delays, speed and current limits provide authentic representations of control surface and motors during the flight envelope. The observed oscillatory responses in the angular rates (p, q, r), body linear velocities (u, v, w) reflect the expected behaviour of the open-loop system in the absence of stabilising feedback control.

Integration of atmospheric turbulence through the discrete-time Dryden model successfully introduces realistic disturbances while maintaining numerical stability via bilinear transformation. Turbulence forces and moments provide essential environmental excitation for a complete validation of the control system under operational conditions.

The multi-sensor architecture implements a sophisticated dual-noise error model capturing both systematic bias drift through Gauss-Markov processes and high-frequency measurement noise. The integration of the IMU model with realistic transfer function dynamics, scale factor errors, and time delays provides authentic measurement data.

This framework provides a robust testbed for advanced research in flight control design, sensor fusion, enabling rigorous evaluation of control architectures under realistic operational conditions such as high-fidelity dynamics, realistic actuator constraints and environmental disturbances, etc..

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